INCH-POUND
MIL-PRF-39001/6A
28 August 1997
SUPERSEDING
MIL-C-39001/6
28 April 1978

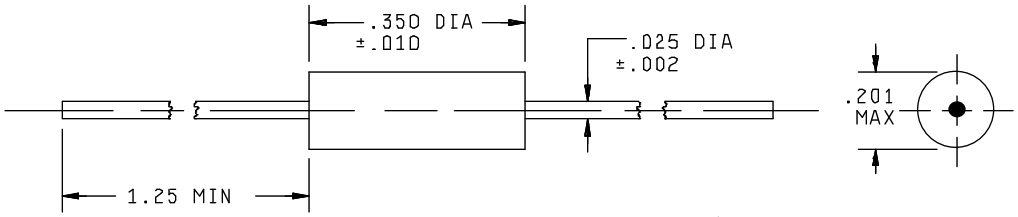
PERFORMANCE SPECIFICATION SHEET

CAPACITORS, FIXED, MICA DIELECTRIC, ESTABLISHED RELIABILITY, STYLE CMR16

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-39001.

FAILURE RATE LEVEL 'L' IS INACTIVE FOR DESIGN AFTER 29 JULY 1983



CIRCUIT DIAGRAM

INCHES	mm
.002	0.05
.010	0.25
.025	0.64
.201	5.11
.350	8.89
1.25	31.8

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Style CMR16 capacitors.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Qualification inspection is not required.

Dimensions: See figure 1.

Case: Tubular, molded plastic, encapsulated.

Capacitance value: See table I.

Capacitance tolerance: See table I.

DC rated voltage: 100V, 300V, or 500 V, as shown in table I.

Operating temperature range: -55°C to +125°C.

Characteristic: C, E, or F, See table I.

Failure rate level: M, P, R, and S, see table I.

Barometric pressure: Method 105, condition D (100,000 ft), of MIL-STD-202. Test potential: 150 percent of rated voltage, not to exceed 500 V dc.

Insulation resistance (IR): Method 302, test condition A, of MIL-STD-202: (Minimum requirements):

At 25°C: 100 Gigaohms
At 125°C: 10 Gigaohms

Dissipation factor (DF): See MIL-PRF-39001.

High voltage stabilization: 200 percent of rated dc voltage for 48 hours at a temperature of 125°C +5°C, -0°C.

Thermal shock and immersion: In accordance with MIL-PRF-39001 except ∆cap.

Moisture resistance: In accordance with MIL-PRF-39001 except ∆Cap. shall not exceed ±1.5 percent of initial measurement or 1pF, whichever is greater.

Life:

Forty eight hours, minimum, at -55°C +0°C -2°C immediately followed by qualification or quality conformance test, as applicable.

Qualification test: 2,000 hours at +125°C with 100 percent or 150 percent of rated dc voltage applied, as applicable.

DWV	-	Meet initial requirement.
IR	-	Meet initial requirement.
∆Cap.	-	Shall not exceed ±1.5 percent of initial measurement or 1 pF, whichever is greater.
DF	-	Not greater than 150 percent of initial requirement.

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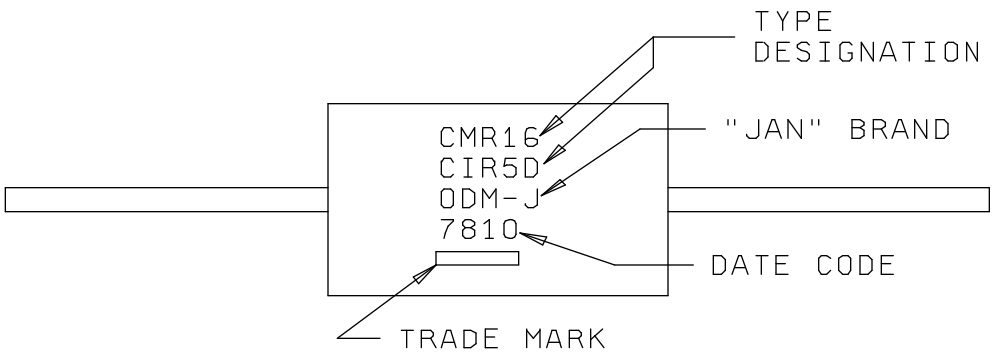

FIGURE 2. Marking examples.

TABLE I. Type designations

STYLE CMR16			
Type designation 1/	Capacitance value	DC rated voltage	Capacitance tolerance available
	(pF)	(volts)	
CMR16C1R0D0D-	1.0	500	D
CMR16C1R5D0D-	1.5	500	D
CMR16C2R0D0D-	2.0	500	D
CMR16C2R5D0D-	2.5	500	D
CMR16C6R0D0D-	6.0	500	D
CMR16C6R5D0D-	6.5	500	D
CMR16C4R0D0D-	4.0	500	D
CMR16C4R5D0D-	4.5	500	D
CMR16C5R0D0D-	5.0	500	D
CMR16C6R0D0D-	6.0	500	D
CMR16C7R0D0D-	7.0	500	D
CMR16C8R0D0D-	8.0	500	D
CMR16C9R0D0D-	9.0	500	D
CMR16C100D0D-	10	500	D
CMR16C110-0D-	11	500	D, J
CMR16C120-0D-	12	500	D, J
CMR16C150-0D-	15	500	D, J
CMR16C180-0D-	18	500	D, J
CMR16E200-0D-	20	500	D, J
CMR16E220-0D-	22	500	D, J
CMR16E240-0D-	24	500	D, J
CMR16E270-0D-	27	500	G, J
CMR16E600-0D-	60	500	G, J
CMR16E660-0D-	66	500	G, J
CMR16E660-0D-	66	500	G, J
CMR16E690-0D-	69	500	G, J
CMR16E460-0D-	46	500	G, J
CMR16E470-0D-	47	500	G, J

See footnote at end of table.

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TABLE I. Type designations - Continued.

STYLE CMR16			
Type designation 1/	Capacitance value	DC rated voltage	Capacitance tolerance available
	(pF)	(volts)	
CMR16E510-0D-	51	500	F, G, J
CMR16E560-0D-	56	500	F, G, J
CMR16E620-0D-	62	500	F, G, J
CMR16E680-0D-	68	500	F, G, J
CMR16E750-0D-	75	500	F, G, J
CMR16E820-0D-	82	500	F, G, J
CMR16F910-0D-	91	500	F, G, J
CMR16F101-0D-	100	500	F, G, J
CMR16F111-0D-	110	500	F, G, J
CMR16F121-0D-	120	500	F, G, J
CMR16F161-0D-	160	500	F, G, J
CMR16F151-0D-	150	500	F, G, J
CMR16F161-0D-	160	500	F, G, J
CMR16F181-0D-	180	500	F, G, J
CMR16F201-0D-	200	500	F, G, J
CMR16F221-0D-	220	500	F, G, J
CMR16F241-0D-	240	500	F, G, J
CMR16F271-0C-	270	600	F, G, J
CMR16F601-0C-	600	600	F, G, J
CMR16F661-0C-	660	600	F, G, J
CMR16F661-0C-	660	600	F, G, J
CMR16F691-0C-	690	600	F, G, J
CMR16F461-0A-	460	100	F, G, J
CMR16F471-0A-	470	100	F, G, J
CMR16F511-0A-	510	100	F, G, J
CMR16F561-0A-	560	100	F, G, J
CMR16F621-0A-	620	100	F, G, J
CMR16F681-0A-	680	100	F, G, J

1/ Complete type designation will include additional letter symbols to indicate the following, where applicable:

Capacitance tolerance: D (±.5 pF); F (±1%); G (±2%); or J (±5%).
Failure rate level: M (1.0%); P (0.1%); R (0.01%); or S (0.001%).

Quality conformance test: 2,000 hours (M, P, R, and S FRLs) at +125°C and 150 percent of rated dc voltage; requirements as specified for qualification.

Resistance to soldering heat: In accordance with MIL-PRF-39001 except ΔCap. shall not exceed ±1.5 percent of initial measurement or 1 pF, whichever is greater.

Marking: Capacitors shall be marked as indicated in figure 2.

Packaging:

Capacitors will be furnished in tape and reel packaging when so specified in the ordering data.

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Application Notes:

Where applicable, these capacitors can be used as replacements for CM15 capacitors listed in MIL-C-5/1.

Substitutability for failure rate and capacitance tolerances shall be as follows:

Failure rate	Will replace failure rate	Cap tolerance	Will replace tolerance
S	R, P, M, L	D	J
R	P, M, L	F	G, J
P	M, L	G	J
M	L		

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Air Force - 19, 99
Navy - MC
DLA - CC

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5910-1945-02)

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